Name _____ Student ID _____

## Part I: Multiple Choice (20 points)

Circle the one correct answer in each question.

1. What is a trap in an operating system?

   (a) A trap is a system call that provides OS level service
   (b) A trap is a service that user can capture output from a process
   (c) A trap is a system call that captures users in OS
   (d) A trap is a way to capture bugs
   (e) There is no trap in OS

2. What protection risk arises if user mode and kernel mode are not separated?

   (a) Process cannot be kicked of processor
   (b) OS cannot access disks or external devices
   (c) Process may be able to access other process data
   (d) OS can prevent process from directly access devices
   (e) Process cannot create child process

3. Which set gives the four components of a process in the model used in class?

   (a) Program
   (b) Data state
   (c) Process status
   (d) Control state
   (e) All of the choices are process components

4. Which OS goal is specifically intended to protect one application from another?

   (a) Performance
   (b) Isolation
   (c) Reliability
   (d) Mobility
   (e) All of the issues is for protection

5. Which low-level hardware mechanism is needed in a preemptive system to switch a CPU-bound process?

   (a) MMU
   (b) Timer interrupt
   (c) Cache
   (d) TLB
   (e) None of them

6. Which part of a process is stored in kernel space?

   (a) Text segment
   (b) Data segment
   (c) Stack segment
   (d) Process descriptor
   (e) All of them are not in kernel

7. In a conventional process address space, which region grows in the opposite direction from the heap?

   (a) Text
   (b) Heap
   (c) Stack
   (d) Program
   (e) They are all growing in the same direction

8. When would a process in the ready state be moved directly to the blocked state?

   (a) When it read a file
   (b) When it read from memory
   (c) When timer interrupt occurs
   (d) Cannot happen
   (e) Both when read from file and memory

9. Which event can move a process into the ready state?

   (a) Timer interrupt occurs
   (b) I/O transfer is completed
   (c) A process is created
   (d) All of the events
   (e) None of the events

10. What does it mean for a process to be a zombie?

    (a) Process is running on processor and cannot be kicked off
    (b) Process is waiting for I/O indefinitely
    (c) Process is ready but has never be select to run
    (d) Process is terminated but cannot be removed from the system
    (e) There is no such thing as zombie in OS

11. Which process-control block stores a process's state and saved CPU context?

    (a) PCB
    (b) TLB
    (c) Stack
    (d) MMU
    (e) Queue

12. Which situation can leave a zombie process in the system?

    (a) Parent exists before child returns value
    (b) Parent creates more than one child
    (c) Child also creates a child process
    (d) Parent waits for child
    (e) Child executes other program

13. In the lab, when a parent waits for the child before printing, what will always happen?

    (a) Child will always print after parent
    (b) A zombie will always happen
    (c) Child will always start from beginning of main function
    (d) A segmentation fault will always occur
    (e) Parent will always print after child

14. The following C program:

```
main()
{
   fork();
   fork();
   printf("hello");
}
```

    How may time the word "hello" will be printed

    (a) Only once
    (b) Twice
    (c) Three times
    (d) Four times
    (e) Eight times

15. Which events can cause a running process to give up the CPU?

    (a) Timer interrupt
    (b) Calling I/O transfer
    (c) Calling wait
    (d) Calling yield
    (e) All of the events

16. Which scheduling algorithm executes jobs to completion in the order of arrival?

    (a) FIFO
    (b) SJF
    (c) RR
    (d) Both FIFO and RR
    (e) Both FIFO and SJF

17. In the C storage-class model covered in class, which class requests register storage for a local variable?

    (a) Auto
    (b) Register
    (c) Extern
    (d) Static
    (e) They are executed at the same speed by CPU

18. Which statement about partition allocation is true?

    (a) Single partition allocation is more efficient that multiple partition allocation in terms of memory usage
    (b) Flexible-size partition does not suffer external fragmentation
    (c) Fixed-size partition suffers from internal fragmentation
    (d) To solve internal fragmentation, OS needs to do memory coalescing and compaction
    (e) None of them is true

19. What does the short-term process scheduler do?

    (a) Select program to load into memory
    (b) Select process to run
    (c) Select process to remove from memory
    (d) Select memory to assign to a process
    (e) Select process to put to memory

20. What does a page-replacement algorithm do when a frame is needed?

    (a) Select a page to load into memory
    (b) Select a page to run
    (c) Select a page to remove from memory
    (d) Select memory to assign to a page
    (e) Select a page to replace a process

21. What is not used in address translation?

    (a) MMU
    (b) Page table
    (c) TLB
    (d) Segment table
    (e) All of them are used in address translation

22. Which option names the fragmentation problem that can occur with contiguous segment allocation?

    (a) Harder to translate from VA to PA
    (b) Cannot protect memory from other process
    (c) Suffer from external fragmentation
    (d) Use too much memory to keep map of the address translation
    (e) There is no disadvantage

23. What does demand paging mean?

    (a) Page will be loaded when it is accesses only
    (b) Page is always in memory for on demand access
    (c) Page is pre-loaded on user demand
    (d) Page is pre-loaded by probability of the on demand usage
    (e) There is no such thing as on demand paging in OS

24. Which step does NOT occur while handling a valid page fault?

    (a) Save process context
    (b) Free a frame if there is no free frame
    (c) Locate and read page from disk
    (d) Update page table
    (e) All events happen

25. Which segment-table field stores a segment's starting physical address?

    (a) Base
    (b) Bound
    (c) Direction
    (d) Valid
    (e) All choices are a part of segment table

26. With a fixed virtual address-space size, what is one effect of using a larger page size?

    (a) More problem with external fragmentation
    (b) Smaller TLB reaches
    (c) Smaller page table
    (d) More process can be created
    (e) There is no effect from the size of page

27. Which single-queue policy rotates among all ready jobs, giving each a time slice in turn?

    (a) FIFO
    (b) STCF
    (c) RR
    (d) MLFQ
    (e) All of the schedulers have the same fairness

28. Which scheduler does not need timer interrupt?

    (a) FIFO
    (b) STCF
    (c) RR
    (d) MLPQ
    (e) All of the schedulers have the same fairness

29. Where does virtual memory save the image of memory?

    (a) On virtual storage
    (b) On a virtual cloud
    (c) On virtual private network
    (d) On kernel of the OS
    (e) On secondary storage

30. What is the response time of a process?

    (a) Time that process starts running until finish
    (b) Time that process enters system until finish
    (c) Time that process enters system until running
    (d) Time that process runs until requests for I/O
    (e) Time that process runs at each cycle

31. What is the turnaround time of a process?

    (a) Time that process starts running until finish
    (b) Time that process enters system until finish
    (c) Time that process enters system until running
    (d) Time that process runs until requests for I/O
    (e) Time that process runs at each cycle

32. What operation may be needed to address external fragmentation from flexible-size partitions?

    (a) Limit degree of multitasking
    (b) Internal fragmentation
    (c) Address translation
    (d) Need compacting
    (e) There is no problem

33. With a single-level page table and fixed page size, what is a cost of a larger virtual address space?

    (a) Need more memory to store page table
    (b) Internal fragmentation
    (c) External fragmentation
    (d) TLB reach is smaller
    (e) There is no problem causes by larger address space

34. How can a multilevel page table save memory for a sparse address space?

    (a) Table size is reduced
    (b) Address space is reduced
    (c) Page size is increased
    (d) All of the second level pages are empty
    (e) There is no need to store all the empty second level pages

35. In the simplified state model, which state does a newly created process enter after admission?

    (a) ready
    (b) running
    (c) blocked
    (d) terminated
    (e) empty

36. Which state is a process in while it is executing on the CPU?

    (a) ready
    (b) running
    (c) blocked
    (d) terminated
    (e) empty

37. What fairness risk can SJF create if shorter jobs keep arriving?

    (a) Suffer convoy effect
    (b) Highest average waiting time
    (c) Highest average response time
    (d) Highest average turn around time
    (e) Starvation

38. Which scheduling metric is Round Robin primarily designed to improve?

    (a) Lowest average turn around time
    (b) Lowest average waiting time
    (c) starvation could occur
    (d) Lowest average response time
    (e) highest throughput

39. Which statement about paging is correct?

    (a) Page and page frame can be different in size
    (b) The smaller the page size is, the smaller the page table is
    (c) Larger page transfer as fast as smaller page
    (d) Page is in logical memory, page frame is in physical memory
    (e) Demand paging is when a program can demand preloading of a page

40. Which page-replacement policy is an ideal benchmark but cannot be implemented exactly without knowing future references?

    (a) Optimal
    (b) FIFO
    (c) LRU
    (d) LFU
    (e) All of the answers can be implemented

41. When does a page fault occur?

    (a) When a process access a page that is not valid in page table
    (b) When a process access a page that is not in logical memory
    (c) When a process access a page that is updated in memory
    (d) When a process access a page not updated in memory
    (e) All of the answers cause page fault to occur

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## Part II: Short Answer (10 points)

Answer all questions briefly in the spaces provided. (1 point each.)

1. Write the full names of MLFQ and STCF.

2. What is the difference between a process and a program?

3. Expand MMU and state its address-translation function.

4. Give two examples of CPU state saved during a context switch.

5. Give two types of information stored in a PCB, other than CPU register values.

6. What is an exited child process called before its living parent reaps its exit status?

7. Give one example each of an I/O-intensive task and a CPU-intensive task.

8. What is thrashing in a virtual-memory system?

9. With base = 16 KiB and bound = 4 KiB, is virtual byte address 4200 valid? Why?

10. Expand TLB and explain why a TLB miss need not cause a page fault.

## Part III: Written Questions (20 points)

Answer all three questions in the answer booklet. Show your work.

1. **(8 points)** Consider the following set of processes (jobs), where the run time of the process is given in milliseconds (ms):

| Process | Run Time | Priority |
|---|---|---|
| $P_1$ | 10 | 3 |
| $P_2$ | 1 | 1 |
| $P_3$ | 2 | 3 |
| $P_4$ | 1 | 4 |
| $P_5$ | 5 | 2 |

   The processes are assumed to arrive in the order $P_1$, $P_2$, $P_3$, $P_4$, $P_5$, all at time 0.

   (a) Draw three diagrams illustrating the execution of these processes using FIFO, SJF, and RR scheduling with time slice = 1 ms.

   (b) What is the Turnaround time of each process for each of the schedulers in part (a)?

   (c) What is the Response time of each process for each of the schedulers in part (a)?

   (d) Which of the schedulers results in (i) the minimum average Turnaround time, (ii) the minimum average Response time (taking the average over all processes)?

2. **(4 points)** Explain how a commercial bank handles customer during rush hours with the concept of the Multi-level Feedback Queue (MLFQ), in your opinion. Describe the algorithms and policies used and provide suggestions on how to improve the process with the goal of optimizing the average turnaround time of customers.

3. **(8 points)** Draw three tables show the 3 Frames in physical memory using FIFO, LRU and OPT for page replacement policy (separate table for both policies) and find the Effective Access Time (EAT) of each policies when references strings (list of needed pages) as below:

   7, 0, 1, 3, 0, 3, 2, 4, 2, 7, 0, 3, 2, 1, 2, 0, 1, 4, 0, 1 (20 needed pages)

   Assume that the memory access time is 10 ns and the disk access time is 10 ms.

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